



Universidad
Carlos III
de Madrid

Robot Intelligence

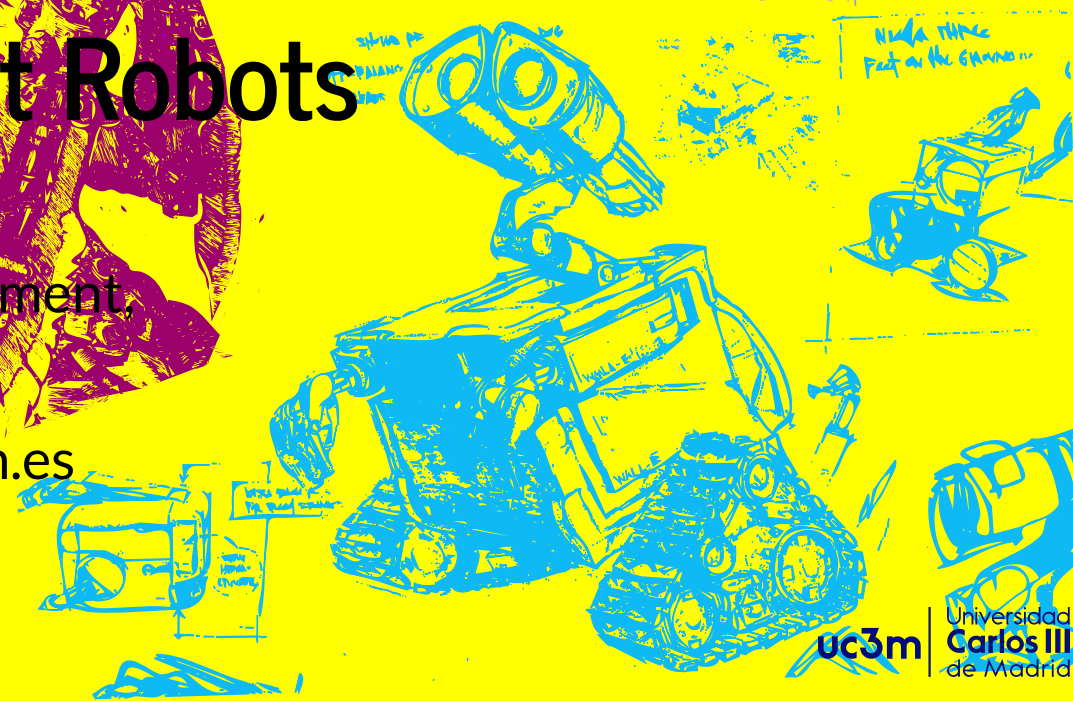
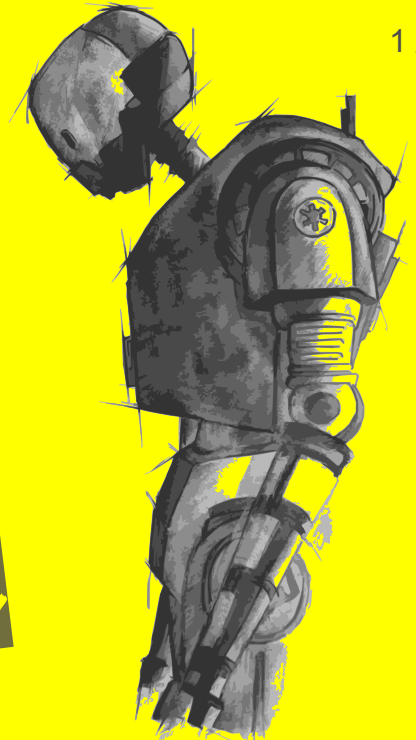
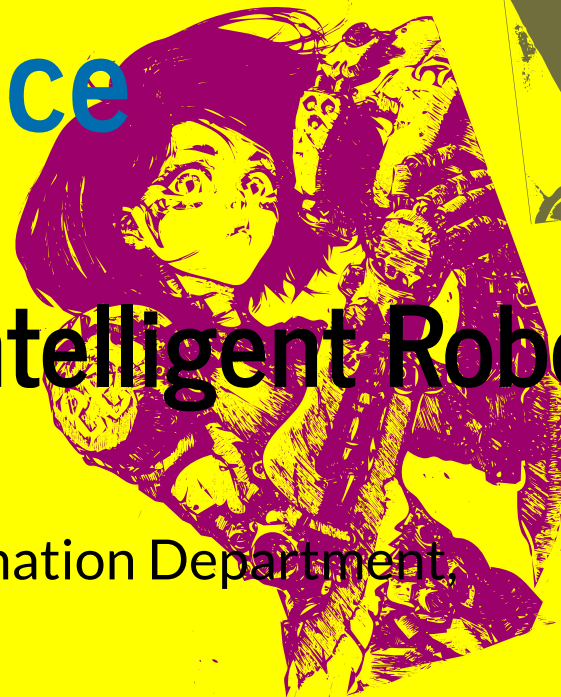
Autonomous and Intelligent Robots

Student: *Sergio Jácono*

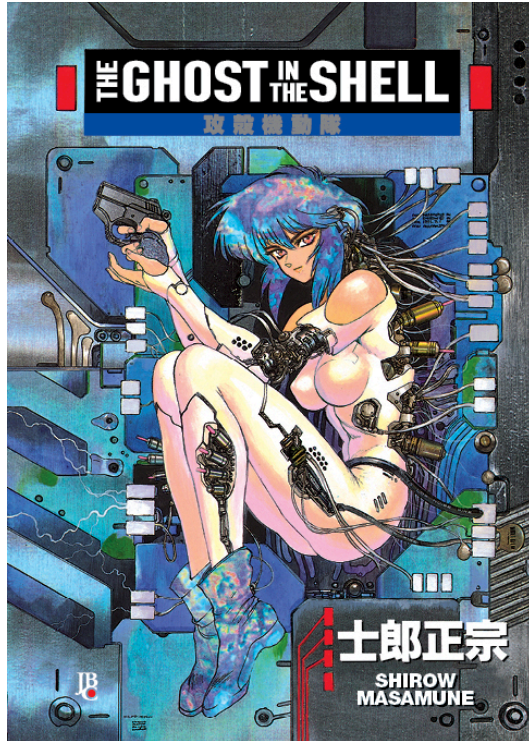
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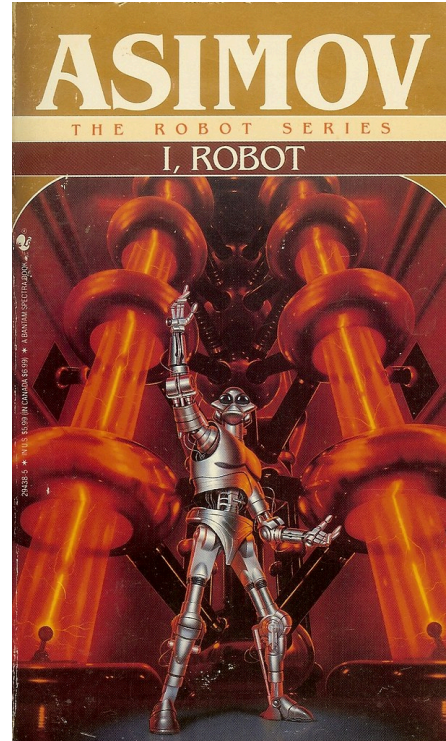
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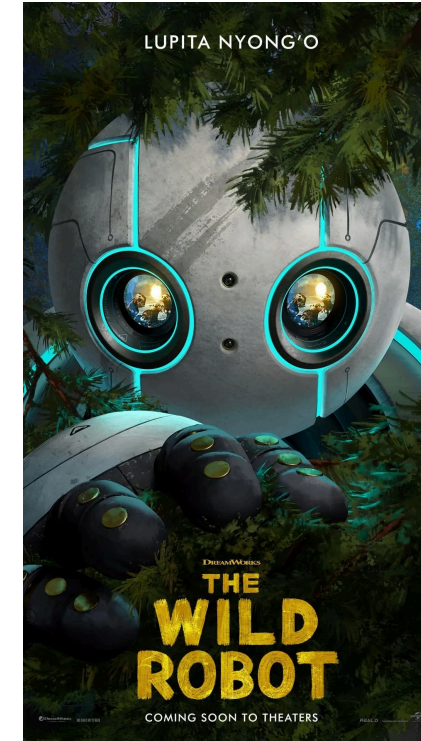
Key ideas



What if a cyber brain could possibly generate its own ghost, and create a soul all by itself? – Motoko Kusanagi



I'm sorry, but you don't understand. These are robots – and that means they are reasoning beings. They recognize the Master, ... They call me the prophet – Cutie (QT-1 robot)



Can I be his mother? – Roz

What does
intelligence mean in
robotics?

A draft meaning



Exploring the surface of an unknown planet is difficult?

The ability to **learn** from its surroundings and previous experiences by performing specific tasks with or without human intervention in a **unknown and complex situations**.

Should a robot be **intelligent**?

Yes, they're becoming more and more intelligent !!!

Why?

- ⦿ Being an autonomous robot also demands a way of learning and a self-improvement in making decision
- 🚫 To work with humans in a more natural, safety and collaborative environment



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Why?

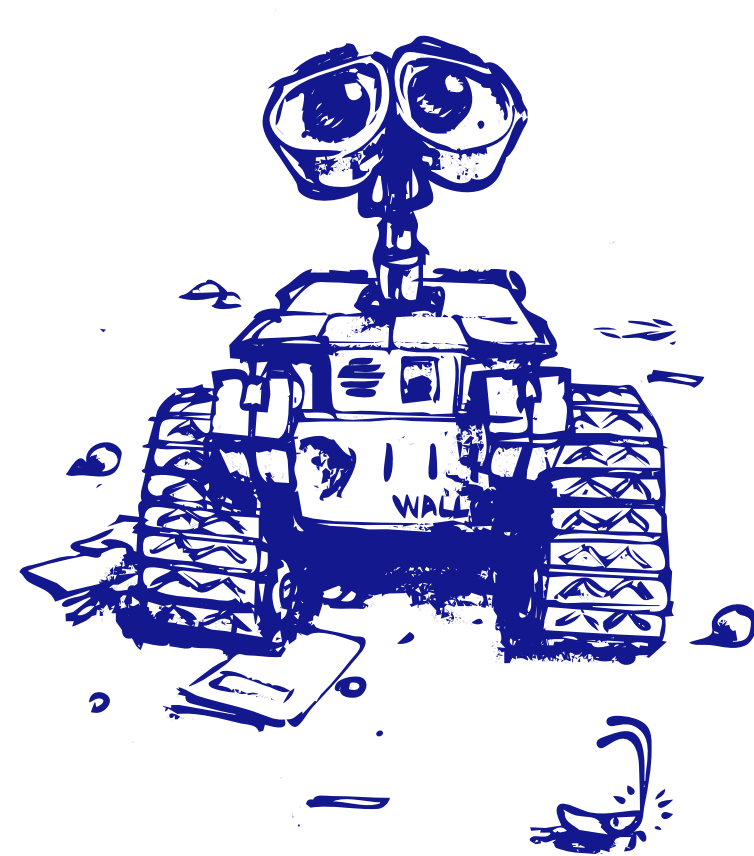
How?

- ⚙ By including perception (such as vision and haptic sensors), object manipulability, recognition and processing systems capabilities.,
- 🤖 Today, Artificial Intelligence can help robots to deal with unexpected situations.



Why we should **treat robotic intelligence differently?**

- Robotics intelligence involves multiples dimensions compared to AI systems in purely digital environments.
- We expect robot have to:
 - Perceive their surroundings
 - Interpret data
 - Act on their decisions by **physically moving**
- Errors in this actions require **fail-safe systems, ethical guidelines, and real-time robustness**



How Much Intelligence Should Be Regularized?

Robots require a balance innovation in

Autonomous decision-making

Interact with humans or operate in public places

Ethical decision-making

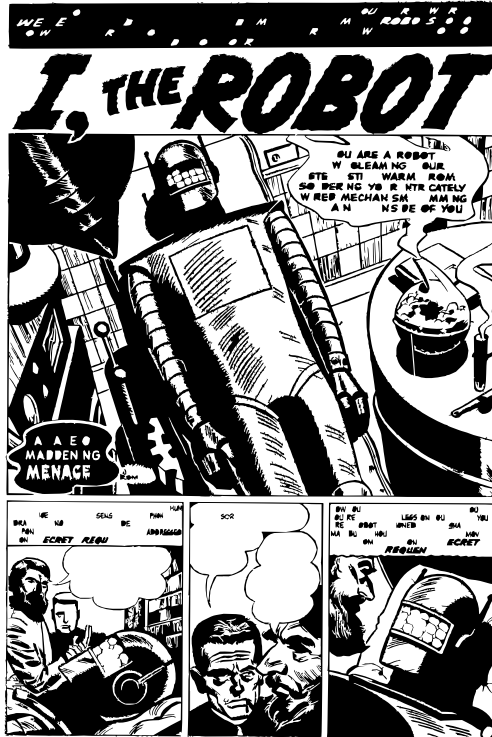
Ensuring that their actions align with human values

Human superviso

Maintain a level of human oversight over robotic decision-making

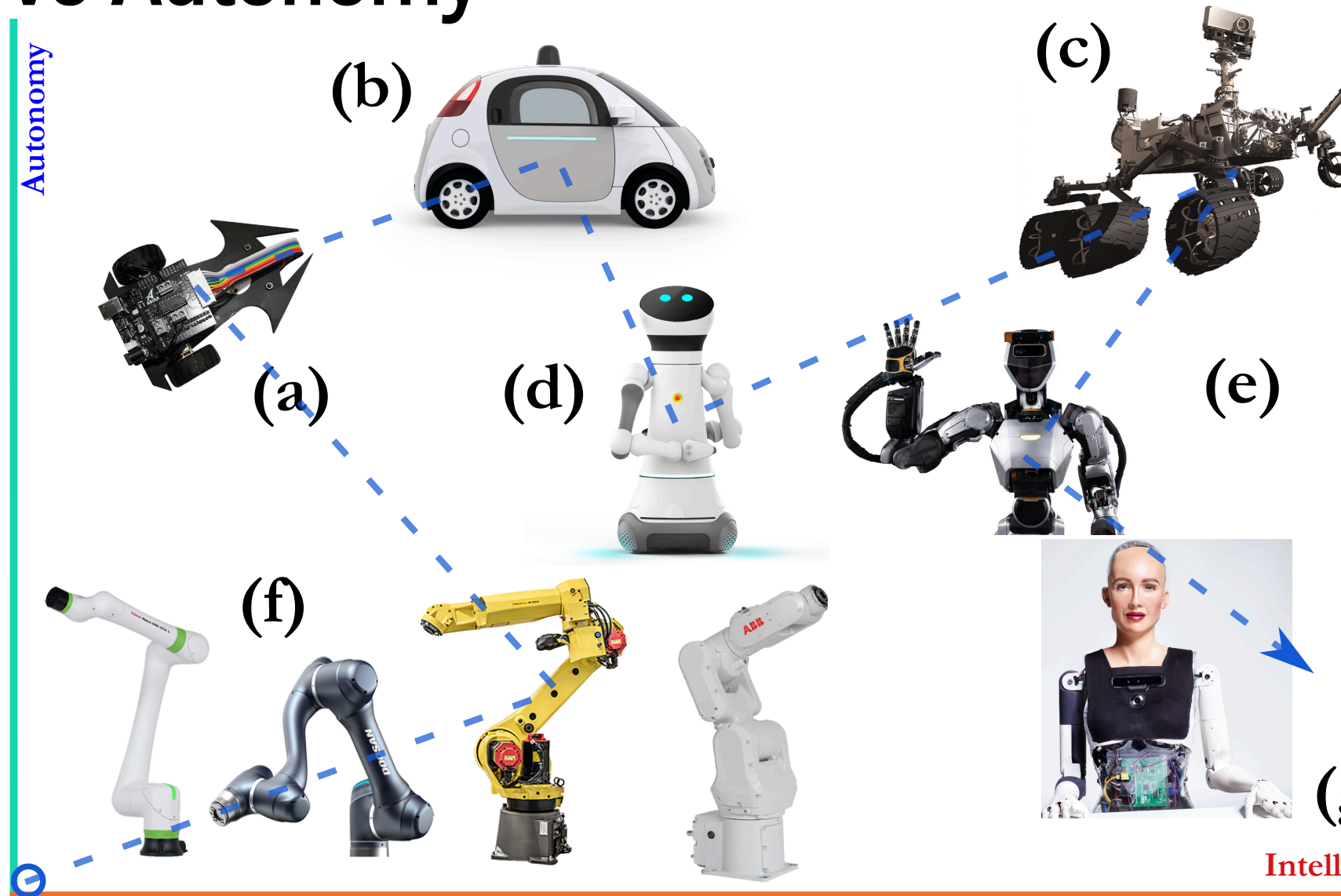
Commercial robots today

Intelligence vs Autonomy

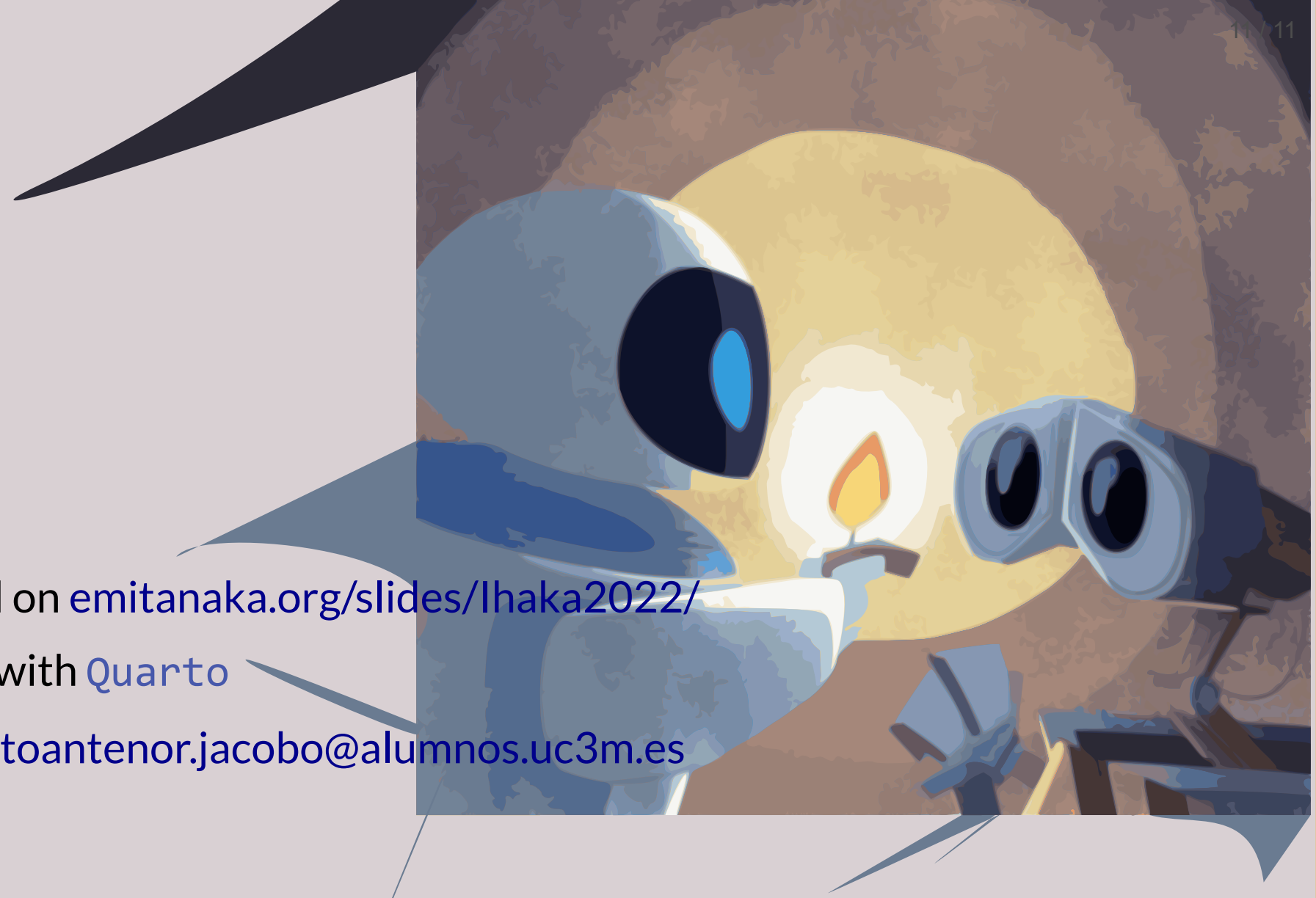


Sci-fi robots vs
Modern robots

Autonomy



Intelligence



🔗 Slides based on emitanaka.org/slides/Ihaka2022/

📄 Created with [Quarto](#)

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